

NOTIFICATION SERVICE - THIET KE CHI TIET

1. Vai tro

Notification Service nhan su kien tu cac service nghiệp vu (chu yeu tu Order Service), luu lich su thong bao theo tung nguoi dung va day thong bao realtime ra giao dien.

2. Luong chi tiet tu dau den dau

Buoc 1: Hanh dong nghiệp vu xay ra trong Order Service

- Vi du: tao don, seller accept/reject, admin ship/deliver.
- Sau khi cap nhat don thanh cong, Order Service tao noi dung thong bao (toUserId, type, title, body, data).

Buoc 2: Order Service gui HTTP request sang Notification Service

- Endpoint nhan: POST /notification-service/publish
- Trong Docker, dia chi noi bo dung:
http://notification-service:4000/notification-service/publish
- Payload chua thong tin:
 - + toUserId: nguoi nhan thong bao
 - + type: loai su kien (ORDER_CREATED, ORDER_STATUS_CHANGED,...)
 - + title, body
 - + data: channel, targetUrl, orderId, shopId, status,...

Buoc 3: Notification Service tiep nhan va dua vao luong xu ly thong bao

- Service kiem tra/normalize du lieu vao.
- Neu cau hinh hang doi bat (RabbitMQ), thong bao duoc day vao exchange/queue.
- Neu tai thoi diem khoi dong broker chua san sang, service retry ket noi dinh ky.

Buoc 4: Notification Service luu thong bao vao Redis

- Moi user co key rieng: notifications:{userId}
- Thong bao moi duoc dua len dau danh sach.
- Service trim so luong item de gioi han kich thuoc du lieu luu.

Buoc 5: Notification Service day realtime den client dang online

- Sau khi xu ly va luu xong, service emit socket event den dung user.
- Neu user dang mo app, giao dien co the cap nhat thong bao gan nhu tuc thi.

Buoc 6: Frontend doc lich su thong bao qua API

- Frontend goi endpoint:
GET /notification-service/read-notifications
- Truyen query: userId, channel, page, limit.
- Service doc Redis, loc theo channel (buyer/seller/admin), phan trang va tra ket qua.

Buoc 7: Frontend dieu huong theo targetUrl

- Moi thong bao chua data.targetUrl.
- Khi user bam vao thong bao, frontend dieu huong den man hinh lien quan (vi du /orders/{orderId} hoac /shop/{shopId}/orders/{orderId}).

3. Vi sao thiet ke nhu vay

- Tach luong thong bao khoi luong dat don giup order response nhanh hon.
- Redis phu hop cho timeline thong bao: append nhanh, doc nhanh, de phan trang.
- Day realtime qua socket giup UX tot hon so voi chi polling.
- Luu thong bao theo userId + channel giup frontend loc dung theo vai tro.

4. Diem can dung khi chay Docker

- Order Service phải set đúng:
NOTIFICATION_SERVICE_URL=http://notification-service:4000/notification-service
- Nếu ở localhost trong container, order sẽ log lỗi "Failed to publish notification: fetch failed".

5. Kiểm tra nhanh khi nghi ngờ? lỗi

- Kiểm tra log order-service có còn lỗi publish không.
- Kiểm tra log notification-service có thông báo đã xử lý/chạy consumer không.
- Kiểm tra endpoint read-notifications tra dữ liệu với userId/channel đúng không.